Applied Animal Behaviour Science 247 (2022) 105566

Contents lists available at ScienceDirect

Applied Animal Behaviour Science

journal homepage: www.elsevier.com/locate/applanim

Accelerometer derived rumination monitoring detects changes in behaviour around parturition

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ARTICLE INFO

Keywords:
Behaviour
Calving
Dystocia
Parturition
Precision livestock management

ABSTRACT

Monitoring cattle for parturition is a labour-intensive task that is often not feasible in extensive environments. On-animal sensors, however, provide a means of remotely monitoring animals, without the labour requirements of traditional observation. This paper explores the use of a rumination algorithm with accelerometer ear tags to observe the changes in rumination around calving. Calving cattle (n = 17) were housed in an intensive calving barn and movement data was collected using tri-axial accelerometer ear tags. A heuristic approach, whereby data was manually visualised to identify ruminating and non-ruminating periods, was used to train a rumination model tailored to each individual animal. Rumination time, as calculated using the rumination model, decreased on 0d and 1d, with a decline in rumination commencing 6 h prior to parturition. These results explore the use of an individualised rumination model for monitoring changes in rumination relative to parturition. Further studies applying this model in various production systems could lead to the development of a universally applicable model for integration with a commercially available sensor system. These commercial tags could subsequently be used to improve the management of cattle around the parturition period, leading to improved productivity and welfare.

Introduction

Monitoring livestock behaviour can provide producers with insight into an animal's overall health, wellbeing, and reproductive or nutritional state (Forbes et al., 2000; Firk et al., 2002; Beauchemin, 2018; Högberg et al., 2021; Thomas et al., 2021). One key behaviour of interest to animal managers is parturition. During periods of impending calving, producers often increase the frequency of animal monitoring in order to provide timely assistance when difficulties arise.

A difficult or prolonged calving, known as dystocia, is an unfortunate outcome of up to 22.6% of calving events (Mee, 2008). It can result in cow and/or calf death, decreased fertility and productivity, morbidity, and a negative impact on calf productivity and welfare up to 30 days following the event (Dematawena and Berger, 1997; Lombard et al., 2007; Barrier and Haskell, 2011). Calves that have experienced dystocia at birth are subject to hypoxic conditions, and are less vigorous and therefore less likely to consume colostrum, through which they gain immunity (Barrier et al., 2012). As such, it is critical that calving is closely monitored by experienced farm personnel to ensure that adequate assistance is available in the event of a dystocia. Early intervention can minimise the effect of dystocia on the calf by reducing the amount of time spent in the birth canal, thus resulting in improved outcomes. Colostrum can also be given soon after birth to ensure the transfer of passive immunity (Robbers et al., 2021). Likewise, early management of dystocic cows can reduce pain, decrease the likelihood of death, and any underlying conditions, injuries, or infections can be treated (Jackson, 2004). The ability to autonomously detect parturition could provide producers with enhanced capacity to manage and mitigate any complications. In more extensive environments, where animals may be physically isolated from the producer and intervention is not possible, parturition detection may act as a management tool for the occurrence of calving and the identification of birth dates.

Current parturition monitoring techniques in dairy or intensive beef industries involve experienced personnel observing cattle for the behavioural or physiological signs of calving, such as vaginal discharge, udder distention, isolation, or increased restlessness (Chang et al.,

Abbreviations: AIC, Akaike information criterion; AR(1), First order autoregressive; ARH(1), First order heterogenous autoregressive; CART, Classification and regression tree; MV, Movement variation.

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https://doi.org/10.1016/j.applanim.2022.105566
Received 23 August 2021; Received in revised form 19 January 2022; Accepted 21 January 2022
Available online 24 January 2022

2020). This method, however, is time consuming, requires experience, and is often not suitable in extensive environments where animals are not regularly monitored during calving (once a day or less). On-animal sensing systems could be used to overcome these limitations by providing a means of remotely and autonomously monitoring cows for the behavioural or physiological changes associated with impending parturition. Several of these devices are commercially available and their use on farm has grown, particularly in intensive systems, however, there are numerous limiting factors, including battery and data transmission complications, that are currently preventing their widespread use (Chapa et al., 2020). The available commercial parturition detection devices fall into three major categories – 1. those designed for intravaginal deployment, that will measure the mechanical separation of the vulval lips or a temperature reduction as the device is expelled; 2. abdominal belts to measure contractions; and 3. a collar, ear, or tail-based accelerometer designed to capture behavioural changes (Saint-Dizier and Chastant-Maillard, 2015; Chang et al., 2020; Miller et al., 2020). Tail-based accelerometers, such as Moocall, have the capacity to detect calving with high accuracy, however, retention issues, injury, and technical problems have been associated with their use (Miller et al., 2020; Giaretta et al., 2021; Pfeiffer et al., 2021). Similarly, the C6 Birth Control system, designed to be sutured into the vulva, provides an accurate detection of calving, however, is most suited to an intensive production system and requires surgical deployment (Paolucci et al., 2008; Marchesi et al., 2013). Consequently, these devices are not appropriate for use in extensive environments.

The commercial tag industry is currently focussed on the development of a device in an ear-tag form factor, given the ease of incorporation into conventional husbandry practises. Previous papers have investigated the use of an ear-tag based device for calving detection, however, these papers have exclusively utilised the CowManager SensOor or Smartbow commercial ear tag systems, both of which contain proprietary rumination algorithms (Pahl et al., 2014; Ouellet et al., 2016; Krieger et al., 2019). This focus on a rumination-based algorithm is likely due to the relationship between rumination and a number of reproductive and welfare states (Calamari et al., 2014; Chang et al., 2022). Therefore, a device capable of measuring and monitoring rumination would have significant value within a commercial production system. The purpose of this study is to evaluate the use of the accelerometer ear-tag based rumination model development methodology described in Chang et al. (2022) to capture the changes in rumination around calving. This methodology involves developing individualised rumination models for use with a classification and regression tree (CART) machine learning approach. Additionally, this study sought to identify how rumination changes in the hours and days around parturition.

Methods and material

All experimental procedures were approved by The Ohio State University Institutional Animal Care and Use Committee (application ID: 2018A00000016) and the Central Queensland University Animal Ethics Committee (application ID: 0000021630). This current study reports on the application of algorithms developed in Chang et al. (2022).

Animals and location

A total of 40 multiparous (average age = 4.2 ± 1.7) Angus crossbreed cows were housed in a semi-enclosed barn (48.77 m × 36.58 m) at The Ohio State University's Don Scott Farm (40°05′N, 83°05'W), 10 km northwest of Columbus, Ohio, United States of America between March 2019 and April 2019. Animals had ad libitum access to alfalfa/grass hay and were supplemented with 50% cracked corn and 50% soyhull concentrate. The concentrate was fed out daily, while the hay was fed at irregular intervals and was replenished as animals finished consuming it (Appendix I). Further details of housing and management procedures can be found in Chang et al. (2022).

Of the 40 cows recruited to this study, 28 calved during the experimental period. One cow gave birth to twins, while another gave birth to a dead calf, but did not experience any dystocia. Two cows experienced calving difficulty and had to be assisted. A total of four animals had incomplete datasets due to power failure of the accelerometer ear tag. Three cows calved prior to camera deployment and as such, the exact calving day and time could not be identified. A total of 17 cows gave birth to live, single calves without assistance, and were utilised for data analysis.

Calving observations

Cattle were continuously monitored via eight surveillance cameras (Swann PRO-T852, Swann Communications, Melbourne, Australia) to identify the exact time of calving. The cameras were in operation between 21 March to 12 April 2019. The video footage was downloaded from the associated DVR unit (Swann DVR-4575, Swann Communications, Melbourne, Australia) using the inbuilt HomeSafe View software. The footage was reviewed by the first author and the time of calving was considered the point at which the hind legs of the calf were completely expelled. All calves in this study were born forward facing.

Accelerometer ear tags

Accelerometer ear tags (AX3 Puck, Axivity Ltd., Newcastle, United Kingdom), calibrated at 12.5 Hz, were attached to the right ear of each cow using an Allflex ear tag applicator (Allflex Australia Pty Ltd., Capalaba, Australia). These tags were deployed for the period spanning 12 March to 15 April 2019.

Data analysis

The raw accelerometer data was downloaded using the AX3/AX6 OMGUI Configuration and Analysis Tool software (Open Movement, Newcastle, United Kingdoms) in .csv format for the period between 21 March to 12 April 2019.

A total of 240 features were generated using the mixed model epoch approach (Chang et al., 2022). This method involved creating 40 metrics for the 1 s, 5 s, 10 s, 30 s, 60 s, and 90 s epochs and joining the data according to timestamp (Table 1). Each epoch was created using a discrete time point and epochs did not overlap. Longer epochs were repeated for each corresponding time period (Chang et al., 2022).

The accelerometer data was divided into the four days prior to and following calving (−4d to 4d) using R statistical software (RStudio Inc., Boston, United States of America). Day 0 was assigned to the calendar day on which calving occurred (00:00:00–23:59:59). The hourly data was also analysed for the 24 h on either side of parturition (−24 to 24 h). Hour 0 referred to the point at which the calf was completely expelled, with Hour 1 commencing 1 h following calving.

Data from a single day (21 March 2019) was quarantined for rumination model development for all animals. This day was selected as it did not fall within the window for anticipated behavioural changes relating to calving and maximised the number of cattle datasets that could be used. The remaining data was used for further analysis. A workflow can be seen in Fig. 1.

Rumination model development

Data from 21 March 2019 was used to develop an individualised rumination model for each animal, with 70% of the data being designated to training the model and 30% of the data utilised for testing the model (Chang et al., 2022). Data was categorised as "ruminating" or "non-ruminating" using a heuristic approach, which took into account animal biomechanics for movement characterisation (Chakravarty et al., 2019). Sporadic video failure and difficulties in distinguishing between

Table 1
Forty features were calculated for each epoch (1 s, 5 s, 10 s, 30 s, 60 s, 90 s) (total = 240 features) using the raw X, Y, and Z accelerometer axes, where t is time and n is the number of counts in the epoch. Equations were adapted from Barwick et al. (2018) and Benaissa et al. (2019).

Feature	Description
Average X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, and entropy(μ_X, μ_Y, μ_Z, μ_{all}, μ_{MI}, μ_{MV}, μ_{energy}, $\mu_{entropy}$)	$\mu_X = \frac{1}{n}\sum_{t=1}^{n} x(t)$ $\mu_Y = \frac{1}{n}\sum_{t=1}^{n} y(t)$ $\mu_Z = \frac{1}{n}\sum_{t=1}^{n} z(t)$ $\mu_{all} = \frac{1}{n}\sum_{t=1}^{n}(x(t)+y(t)+z(t))$ $\mu_{MI} = \frac{1}{n}\sum_{t=1}^{n}\left(\sqrt{x(t)^2+y(t)^2+z(t)^2}\right)$ $\mu_{MV} = \frac{1}{n}\sum_{t=1}^{n}(\lvert x_t - x_{t-1}\rvert + \lvert y_t - y_{t-1}\rvert + \lvert z_t - z_{t-1}\rvert)$ $\mu_{energy} = \frac{1}{n}\sum_{t=1}^{n}(x(t)^2+y(t)^2+z(t)^2)$ $\mu_{entropy} = \frac{1}{n}\sum_{t=1}^{n}\left((1+(x(t)+y(t)+z(t))^2 \times ln\left(1+(x(t)+y(t)+z(t))^2\right)\right)$
Minimum X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, and entropy(min_X, min_Y, min_Z, min_{all}, min_{MI}, min_{MV}, min_{energy}, $min_{entropy}$)	The minimum X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, or entropy value for the epoch
Maximum X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, and entropy(max_X, max_Y, max_Z, max_{all}, max_{MI}, max_{MV}, max_{energy}, $max_{entropy}$)	The maximum X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, or entropy value for the epoch
Standard deviation for X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, and entropy(SD_X, SD_Y, SD_Z, SD_{all}, SD_{MI}, SD_{MV}, SD_{energy}, $SD_{entropy}$)	$SD_X = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(x(t)-\overline{x})^2}$ $SD_Y = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y(t)-\overline{y})^2}$ $SD_Z = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(z(t)-\overline{z})^2}$ $SD_{all} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(((x+y+z)(t))-(\overline{x+y+z}))^2}$ $SD_{MI} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(MI(t)-\overline{MI})^2}$ $SD_{MV} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(MV(t)-\overline{MV})^2}$ $SD_{energy} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(energy(t)-\overline{energy})^2}$ $SD_{entropy} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(entropy(t)-\overline{entropy})^2}$
Range of X axis, Y axis, Z axis, all axes, movement intensity, movement variation, energy, and entropy(R_X, R_Y, R_Z, R_{all}, R_{MI}, R_{MV}, R_{energy}, $R_{entropy}$)	$R_X = max_X - min_X$ $R_Y = max_Y - min_Y$ $R_Z = max_Z - min_Z$ $R_{all} = max_{all} - min_{all}$ $R_{MI} = max_{MI} - min_{MI}$ $R_{MV} = max_{MV} - min_{MV}$ $R_{energy} = max_{energy} - min_{energy}$ $R_{entropy} = max_{entropy} - min_{entropy}$

Where t is the time and n is the total number of counts in the epoch.

different animals made it impossible to visualise rumination for the entirety of a single day.

The mean movement variation (MV) was calculated from the raw accelerometer data using a 1 s epoch in R and was graphed using the R package 'ggplot2' (Wickham et al., 2020). Observation of the ruminating video footage and corresponding MV data revealed a distinctive pattern of acceleration, consisting of a period of activity associated with the chewing portion of rumination, followed by a short decrease in activity corresponding to the bolus being swallowed, before chewing resumes (Fig. 2). The active chewing period usually lasted between 50 and 80 s at an MV between 0.05 and 0.25 (Fig. 2). Similar patterns were also observed using collar mounted accelerometer in cattle (Konka et al., 2014; Davison et al., 2020), ear-tag mounted accelerometers in sheep (Mansbridge et al., 2018), and noseband pressure sensor in cattle (Zehner et al., 2017). Video footage of rumination and the corresponding MV can be found in the supplementary material.

The model development dataset was validated in R. Each individual animal dataset for the model development period was split into 70% training and 30% testing, as per standard data mining procedures (Robert et al., 2009). Using this method, a machine learning model was developed for each animal. This model was developed using 70% of the data and was subsequently trained on the remaining 30% of the data to generate accuracy, sensitivity, specificity, and precision values (Appendix II). This process was repeated 100 times for each animal and each repetition underwent a 10-fold cross validation, as described in Fogarty et al. (2020a). Cross validation involves resampling the data for different training and testing datasets. This method can be utilised to identify issues like overfitting or selection bias, and provides insight into the ability for the model to make predictions on unseen data (Cawley and Talbot, 2010). The rumination model was validated using a CART machine learning approach and the 'rpart' R package (Therneau et al., 2019; Chang et al., 2022). Accuracy, sensitivity, specificity, and precision values for each rumination model are reported in Appendix II.

The individualised rumination models generated using data from 21 March 2019 were used to make behavioural predictions on the remaining accelerometer data (22 March 2019 – 12 April 2019) using a CART (Chang et al., 2022). The proportion of time spent ruminating per day and hour was calculated using these predictions.

Statistical analysis

A linear mixed effects model was developed in R using the 'nlme' package for the ruminating time, as predicted using the CART, for the 4d and 24 h prior to and following calving (−4d to 4d; −24 to 24 h) (Pinheiro et al., 2020). The day or hour around calving and days relative to a supplementary feeding day were treated as fixed effects, while the individual animals were treated as random effects and were subject to repeated measures (Fogarty et al., 2020b). Feed day was incorporated into the linear mixed effects model as rumination time and feeding time are inversely proportional (Schirmann et al., 2012).

Autoregressive models are utilised in environments where there is some correlation between the values and those that precede or succeed them. First order autoregressive (AR(1)) and first order heterogenous autoregressive (ARH(1)) structures were compared, with the final model being developed on the structure with the lowest Akaike information criterion (AIC) score (Fogarty et al., 2020b). The AIC score is used to estimate the quality of the statistical model, with lower values resulting in a higher quality model (Mohammed et al., 2015).

Pairwise comparisons with Tukey adjustment and least squares means were generated for the day and hour models using the 'lsmeans' package (Lenth, 2018; Fogarty et al., 2020b). The upper and lower 95% confidence intervals were also calculated.

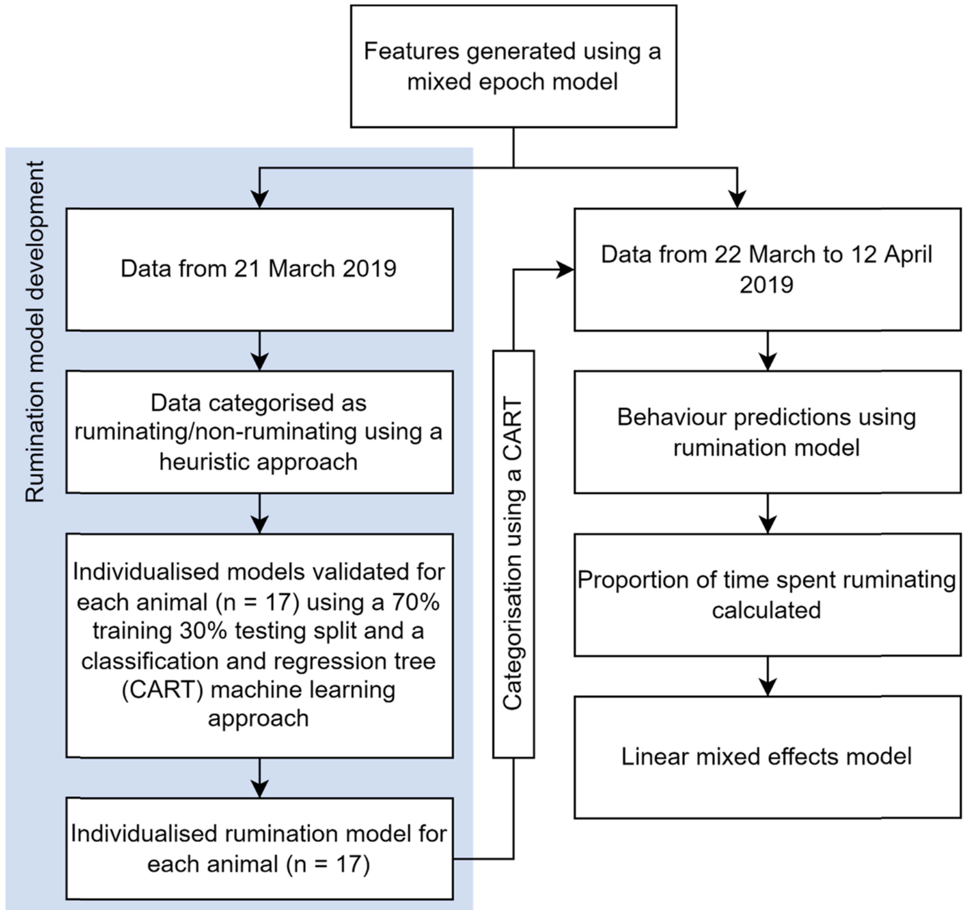

Fig. 1. Data flow for rumination model development and behaviour prediction using rumination model.

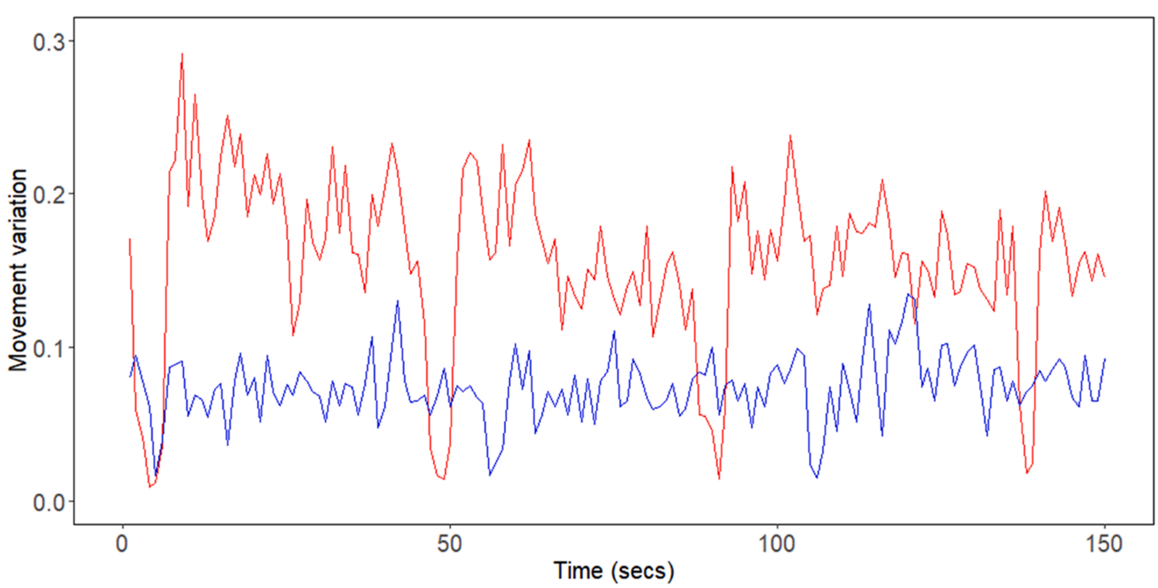

Fig. 2. The mean movement variation using a 1 s epoch for two ruminating animals across 150 s. The rumination signal consists of a period of activity coinciding with chewing, followed by a decline in activity representative of the bolus being swallowed, before activity resumes.

Results

Day

The results of the linear mixed effects model indicate that there was a significant difference in the proportion of time spent ruminating between all days in the prepartum period (−4d to −1d) and the day of calving ($P < 0.0001$) (Fig. 3). The proportion of time spent ruminating decreased to a minimum on the day of calving (21.3%; range: 7.9–34.9%) and remained suppressed on 1d (21.7%; range: 8.8–37.1%) (Fig. 3). The proportion of time spent ruminating was lower post-calving compared to pre-calving (25.6% vs. 31.3%) (Fig. 3).

Hour

The proportion of time spent ruminating in the 24 h prior to calving was significantly different compared to the hour of calving for − 23 h ($P = 0.04$), − 21 h ($P = 0.05$),− 20 h ($P = 0.04$), − 17 h ($P < 0.0001$), − 16 h ($P = 0.004$), − 15 h ($P = 0.002$), − 14 h ($P < 0.0001$), − 13 h ($P = 0.05$), − 12 h ($P < 0.001$), − 11 h ($P = 0.001$), − 10 h ($P = 0.002$), − 9 h ($P = 0.01$), − 8 h ($P = 0.001$), − 7 h ($P < 0.0001$), and − 6 h ($P = 0.02$) (Fig. 4). A decrease in time spent ruminating was observed starting in − 6 h, before reaching a minimum at 0 h, where the observed proportion of time spent ruminating was 1.3% (range: 0.0–9.3%) (Appendix IV). Time spent ruminating was not significantly different to 0 h

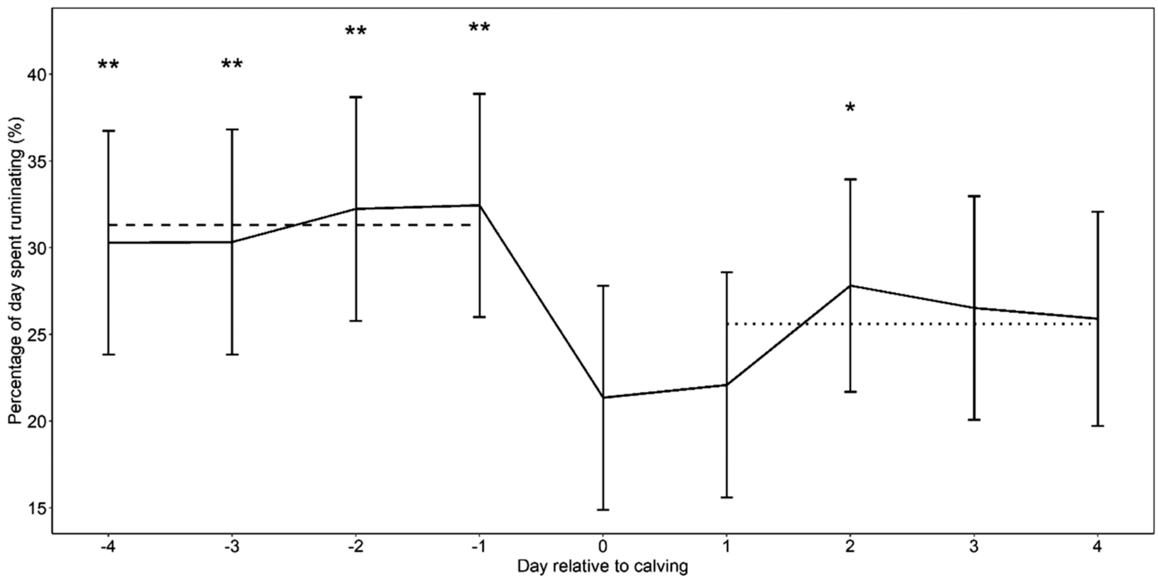

Fig. 3. Proportion of time spent ruminating in the 4d prior to and following calving (n = 17 animals). Error bars represent the 95% confidence interval. Time spent ruminating on days that were significantly different ($P \leq 0.05$) to the calving day are denoted with '* ', whilst days marked with '* *' had a P value < 0.0001 compared to calving day. The dashed line represents the mean proportion of time spent rumination pre-calving, whilst the dotted line represents the mean proportion of time spent rumination post-calving.

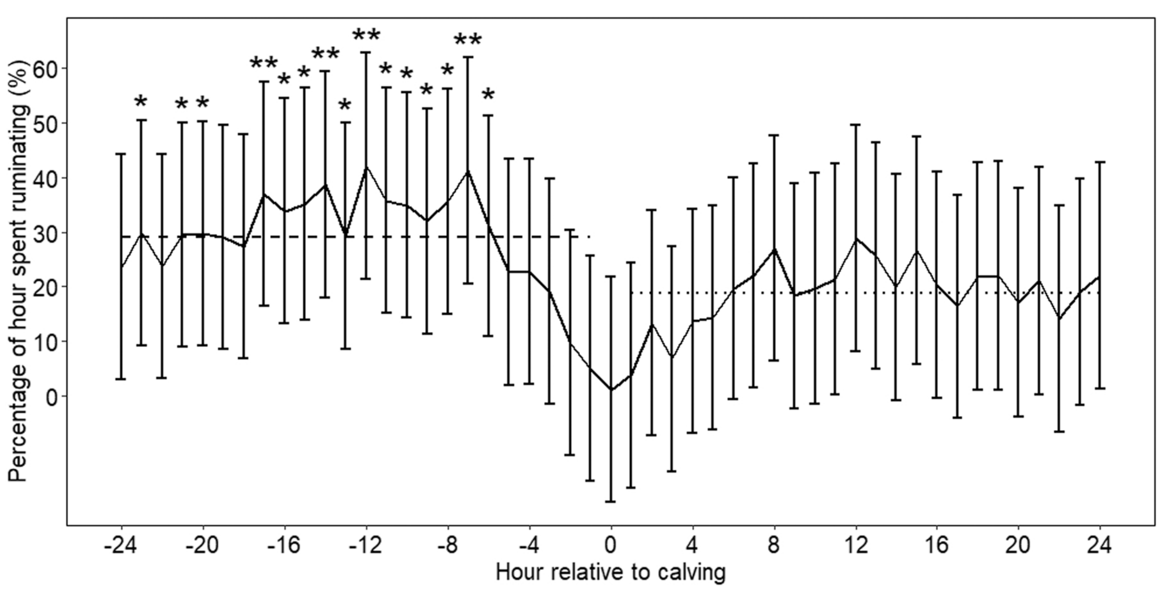

Fig. 4. Proportion of time spent ruminating in the 24 h prior to and following calving (n = 17 animals). Error bars represent the 95% confidence interval. Time spent ruminating for hours that were significantly different ($P \leq 0.05$) to the calving hour are denoted with '*', whilst hours marked with '**' had a P value < 0.0001 compared to calving hour. The dashed line represents the mean proportion of time spent rumination pre-calving, whilst the dotted line represents the mean proportion of time spent rumination post-calving.

for the 24 h following calving (Fig. 4). Eleven of the 17 cows that calved did not ruminate in 0 h.

Discussion

This study describes the use of an accelerometer ear tag for detecting changes in rumination in the hours and days around calving. Previous papers have utilised commercial sensor systems to capture the changes in rumination around calving, however, these systems contain proprietary algorithms. This study describes, for the first time, how a machine learning derived rumination classification model may be used to detect parturition in cattle with accelerometer ear tags.

Rumination model

The rumination modelling process adopted in this study was based on the concept proposed in Chang et al. (2022), which involved developing individualised rumination models for use with a CART machine learning approach. To confirm the overall accuracy of the proposed model, it is worth comparing the overall rumination patterns observed against what has been reported in the literature. Utilising the model described, a baseline average rumination time of 441.3 ± 98.8 min/d was reported. Rumination time, as measured using validated methods in dairy cattle, range from a reported 151 min/d up to 610 min/d (Zebeli et al., 2006; White et al., 2017). Average rumination times were measured at 436 min/d (n = 179 observations, 60 studies) by White et al. (2017) and 434 min/d (n = 24 studies) by Zebeli et al. (2006). Therefore, the reported rumination time in this study was not substantially different compared to rumination times reported in other studies using validated methods.

Rumination variation around calving

The results of this paper indicate that rumination, as measured using an accelerometer ear tag and rumination model, decreased at both a day and subday level relative to parturition. These results are corroborated by findings in a systematic literature review conducted by Chang et al. (2020), which explored the behavioural indicators of calving. In this review, a general trend of decreasing rumination was reported around parturition.

Rumination changes at the day level

Rumination time on the day of calving averaged 306.9 ± 119.1 min/day (range: 113.5 min/day – 502.0 min/day), a 33.9% decrease in comparison to the day prior (Fig. 3). Several papers have reported similar findings for sensor-derived rumination monitoring around parturition. Borchers et al. (2015) utilised an HR-Tag (a collar-based acoustic monitoring device) and reported a 34.6% decrease ($P < 0.05$) in rumination between the day prior to and the day of calving (Table 2). No significant differences were observed between all days pre-partum, a result also noted in this study. Similarly and whilst using the same tag, Schirmann et al. (2013) reported a total rumination time on the day of calving at 293.3 ± 7.9 min, however, this equated to a decrease of 19.2% from the day prior (Table 2).

Table 2
Comparison of rumination time observed using sensors on a minutes per day basis of this study to other papers in the 4 days prior to and following calving.

Day	This study	Borchers et al. (2015)	Pahl et al. (2014)	Schirmann et al. (2013)
-4	441.3 ± 98.8	349.2 ± 13.7		
-3	422.6 ± 76.0	344.8 ± 13.7		
-2	461.0 ± 110.0	332.0 ± 13.7		
-1	464.1 ± 93.4	324.1 ± 13.7	374 ± 110	362.8 ± 7.6
0	306.9 ± 119.1	211.9 ± 13.6	278 ± 64	293.3 ± 7.9
1	311.9 ± 93.4			428.9 ± 26.5
2	393.5 ± 98.4			
3	384.5 ± 113.4			
4	365.9 ± 118.2			

The results of this study indicate that rumination remained suppressed on the day after calving. Schirmann et al. (2013) has been the only study to explore this and reported the same findings, with rumination returning to baseline levels in the 24–48 h following calving.

It should be noted that the described studies measured rumination in dairy cattle. Differences in cattle breed can significantly impact upon time spent ruminating (Carvalho et al., 2020). As such, additional studies are required to evaluate the described model in dairy cattle.

Rumination change at an hourly level

The findings from this study suggest that there is a decline in rumination commencing 6 h prior to parturition. Findings by Houwing et al. (1990) further support this result, with an observed decline to 5.7% of the time spent ruminating in the 6 h prior to calving. Likewise, Büchel and Sundrum (2014) reported a reduction in ruminating time 6 h prior to parturition in 88% of the experimental animals. A study conducted by Miller et al. (2020) measured rumination time in beef and dairy cows using neck mounted accelerometers and observed a significant decrease in rumination compared to baseline levels in beef cows commencing 5 h prior to calving. This decline accounted for a decrease in rumination of approximately 50%. In this study, rumination time declined by 27% compared to the recorded baseline value.

Implications for livestock management

There are a number of potential applications of the information presented in this study. For some livestock managers, using sensors might give them the ability to predict the onset of calving and would allow producers to more effectively manage their animals during this critical period. In intensive calving operations, cattle can be more closely monitored and intervention can be initiated if required. The resultant reduction in cow and/or calf loss and the ability to administer postpartum care can improve productivity and animal welfare outcomes. In more extensive operations, operational decisions can be made to potentially move cattle to an environment where they can be observed more closely, such that producers can reap the increased productivity and welfare benefits. Where intervention is not possible, particularly in extensive grazing environments, the benefits might be an improved understanding of the key issues occurring during parturition at a herd level that might enable improved animal management in subsequent calving seasons.

Recommendations for future studies

The development of a rumination model adaptable to numerous environments could be integrated with commercially available tags, however, further studies must first be conducted to validate the described model across various production systems, feed types, parities, and cattle breeds. Further studies are required to evaluate the use of this method of capturing changes in rumination for parturition detection.

A decrease in rumination has been observed in response to differing physiological states in cattle, such as oestrus or disease, as well as differing feed sources and characteristics (Pahl et al., 2015; Beauchemin, 2018; Marchesini et al., 2018). Consequently, a decrease in rumination is not necessarily indicative of a calving cow. Additional studies could investigate the use of the described algorithm in conjunction with other sensors to further verify a calving event. Potential candidates worth exploring might include the rate at which rumination changes in the hours prior to calving or the use of a global navigation satellite system to detect changes in spatial use around parturition.

Additional studies could also examine the use of the described model to distinguish between eutocia and dystocia. A significant and prolonged reduction or even cessation in rumination would be expected in cows experiencing dystocia. The further development of this model to autonomously identify dystocia would have significant contributions to improved productivity and welfare on-farm.

Conclusion

The results of this study indicate that the use of an individualised rumination model with an accelerometer ear tag can be used to detect changes in rumination behaviour in the days and hours around parturition. Additional studies are required to determine if this methodology is sufficient for parturition detection. Given that rumination decreases in relation to many changes in the cow's physiological state, such as disease or oestrus, further studies should be conducted to evaluate whether the combination of this rumination algorithm with other unique calving indicators could be used to improve parturition detection. The described approach provides a promising method in which producers, researchers, and commercial tag companies are able to autonomously detect parturition using an ear-tag sensor.

CRediT authorship contribution statement

Anita Z. Chang: Conceptualization and design of the study, Acquisition of data, Analysis and interpretation of data. **Eloise S. Fogarty**: Assisted with analysis and interpretation of data. **David L. Swain**: Supervision. **Alvaro García-Guerra**: Assisted with the acquisition of data. **Mark G. Trotter**: Assisted with conceptualization of the study and interpretation of data, Supervision, Revising and editing the paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge financial support of Central Queensland University, Meat & Livestock Australia, and Telstra. Thanks also to S. Battista, M. Creager, A. Crist, B. Duran, C. Gasser, C. Golden, G. Fogle, M. Mussard, and E. Rojas-Canadas of The Ohio State University for their assistance with field work, and L.B. Ryall.

Appendix I. Table of feed days and date

Date	Feed day
22/03/2019	Yes
23/03/2019	No
24/03/2019	Yes
25/03/2019	No
26/03/2019	No
27/03/2019	Yes
28/03/2019	No
29/03/2019	Yes
30/03/2019	No
31/03/2019	Yes
01/04/2019	No
02/04/2019	No
03/04/2019	Yes
04/04/2019	No
05/04/2019	No
06/04/2019	Yes
07/04/2019	No
08/04/2019	No
09/04/2019	No
10/04/2019	No
11/04/2019	No
12/04/2019	Yes

Appendix II. Accuracy, sensitivity, specificity, and precision values for validation of training dataset generated on data gathered on 21 March 2019

ID	Accuracy	Sensitivity	Specificity	Precision
1	96.1	96.8	94.2	97.8
2	97.4	98.4	95.2	97.8
3	96.5	96.5	96.5	98.5
4	97.2	98.6	93.6	97.4
5	97.5	98.1	96.3	98.0
6	95.4	96.7	93.3	96.0
7	96.0	98.6	90.3	95.8
8	97.8	99.1	95.3	97.7
9	97.6	98.7	95.3	97.7
10	96.9	98.0	94.7	97.3
11	97.8	98.6	96.3	98.0
12	96.9	97.1	96.5	98.1
13	97.2	99.0	94.0	96.9
14	97.4	98.6	94.5	97.6
15	95.9	98.8	90.2	95.2
16	96.5	99.6	90.8	95.2
17	97.5	98.1	96.6	97.7

Appendix III. Pearson correlation table detailing P values for 4d prior to and following calving. Significant differences were declared where P ≤ 0.05 and are denoted with ‘* ’, whilst days marked with ‘* *’ had a P value ≤ 0.001 and days marked with ‘*’ had a P value ≤ 0.0001**

Day	-4	-3	-2	-1	0	1	2	3	4
-4		1.00	0.98	0.97	***	**	0.93	0.56	0.37
-3			0.96	0.97	***	**	0.93	0.56	0.35
-2				1.00	***	***	0.35	0.08	*
-1					***	***	0.29	0.06	*
0						1.00	*	0.14	0.31
1							*	0.28	0.54
2								1.00	0.98
3									1.00
4									

Pearson correlation table detailing P values for 24 h prior to and following calving. Significant differences were declared where P ≤ 0.05 and are denoted with ‘* ’, whilst days marked with ‘* *’ had a P value ≤ 0.001 and days marked with ‘* ** ’ had a P value ≤ 0.0001 (continued on next page).

Hour	-24	-23	-22	-21	-20	-19	-18	-17	-16	-15	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
-24		1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.89	1.00	1.00	1.00	1.00	0.94	1.00	1.00	1.00	1.00	1.00	0.88
-23			1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.73	0.21
-22				1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.90	1.00	1.00	1.00	1.00	0.94	1.00	1.00	1.00	1.00	1.00	0.87
-21					1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.78	0.24
-20						1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.75	0.22
-19							1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.82	0.28
-18								1.00	1.00	1.00	1.00	1.00	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.94	0.46
-17									1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.08	*
-16										1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.27	*
-15											1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.20	*
-14												1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.79	*	*
-13													1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.26
-12														1.00	1.00	1.00	1.00	1.00	1.00	0.81	0.81	0.39	*	***
-11															1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.13	*
-10																1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.19	*
-9																	1.00	1.00	1.00	1.00	1.00	1.00	0.47	0.08
-8																		1.00	1.00	1.00	1.00	0.98	0.14	*
-7																			1.00	0.86	0.88	0.49	*	***
-6																				1.00	1.00	1.00	0.56	0.11
-5																					1.00	1.00	1.00	0.93
-4																						1.00	1.00	0.93
-3																							1.00	1.00
-2																								1.00
-1																								
0																								
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Pearson correlation table detailing P values for 24 h prior to and following calving. Significant differences were declared where P ≤ 0.05 and are denoted with '*', whilst days marked with '**' had a P value ≤ 0.001 and days marked with '***' had a P value ≤ 0.0001 (continued from previous page).

Hour	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
-24	0.46	0.77	1.00	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-23	*	0.13	0.97	0.39	0.98	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00
-22	0.44	0.76	1.00	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-21	*	0.15	0.98	0.43	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00
-20	*	0.14	0.98	0.40	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00
-19	0.06	0.19	0.99	0.49	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-18	0.12	0.33	1.00	0.69	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-17	* **	*	0.32	*	0.36	0.43	0.94	1.00	1.00	0.88	0.96	0.99	1.00	1.00	0.96	1.00	0.97	0.66	1.00	1.00	0.77	0.99	0.40	0.92	1.00
-16	*	*	0.68	0.08	0.73	0.80	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.94	1.00	1.00	0.97	1.00	0.77	1.00	1.00
-15	*	*	0.57	0.06	0.61	0.69	1.00	1.00	1.00	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.87	1.00	1.00	0.93	1.00	0.66	0.99	1.00
-14	* **	* *	0.17	* *	0.20	0.25	0.83	0.97	1.00	0.71	0.87	0.96	1.00	1.00	0.87	1.00	0.90	0.45	0.97	0.97	0.56	0.94	0.23	0.78	0.97
-13	*	0.17	0.99	0.46	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00
-12	* **	* **	*	* *	*	0.06	0.43	0.74	1.00	0.31	0.49	0.70	1.00	0.98	0.48	1.00	0.54	0.14	0.72	0.74	0.20	0.64	*	0.38	0.73
-11	*	*	0.45	*	0.50	0.58	0.98	1.00	1.00	0.95	0.99	1.00	1.00	1.00	0.99	1.00	0.99	0.79	1.00	1.00	0.87	1.00	0.54	0.97	1.00
-10	*	*	0.56	*	0.61	0.69	0.99	1.00	1.00	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.88	1.00	1.00	0.93	1.00	0.66	0.99	1.00
-9	*	*	0.87	0.18	0.90	0.93	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	0.92	1.00	1.00
-8	*	*	0.46	*	0.51	0.59	0.98	1.00	1.00	0.95	0.99	1.00	1.00	1.00	0.99	1.00	0.99	0.81	1.00	1.00	0.88	1.00	0.56	0.97	1.00
-7	* **	* **	0.06	* *	0.07	0.09	0.54	0.83	1.00	0.40	0.60	0.79	1.00	1.00	0.59	1.00	0.65	0.19	0.81	0.83	0.27	0.74	0.08	0.48	0.82
-6	*	0.07	0.92	0.24	0.94	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.96	1.00	1.00
-5	0.57	0.86	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-4	0.56	0.86	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-3	0.92	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.95	1.00	1.00	1.00	0.833	0.99	1.00	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.96	0.50	1.00	1.00	0.98	0.30	0.68	1.00	0.55	0.99	1.00	0.97	0.96	1.00	0.98	1.00	1.00	0.96
0		1.00	1.00	1.00	1.00	1.00	0.87	0.65	0.14	0.96	0.90	0.76	0.06	0.25	0.87	0.17	0.83	0.99	0.67	0.66	0.99	0.76	1.00	0.93	0.66
1			1.00	1.00	1.00	1.00	0.99	0.91	0.37	1.00	0.99	0.95	0.20	0.54	0.98	0.41	0.98	1.00	0.92	0.91	1.00	0.95	1.00	1.00	0.91
2				1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
3					1.00	1.00	1.00	0.99	0.72	1.00	1.00	1.00	0.50	0.87	1.00	0.77	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00
4						1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
5							1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
6								1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
7									1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
8										1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
9											1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
10												1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
11													1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
12														1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
13															1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
14																1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
15																	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
16																		1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
17																			1.00	1.00	1.00	1.00	1.00	1.00	1.00
18																				1.00	1.00	1.00	1.00	1.00	1.00
19																					1.00	1.00	1.00	1.00	1.00
20																						1.00	1.00	1.00	1.00
21																							1.00	1.00	1.00
22																								1.00	1.00
23																									1.00
24																									

Appendix IV. Time spent ruminating predicted by rumination model in minutes per day, as detected by sensor

ID	-4	-3	-2	-1	0	1	2	3	4
1	261.5	340.0	239.0	367.5	115.0	270.0	381.0	297.0	204.0
2	447.0	442.5	388.5	460.5	312.0	339.0	465.0	375.0	294.0
3	343.5	445.5	524.5	431.0	369.5	303.0	427.0	560.5	386.0
4	409.5	342.0	475.5	472.5	442.5	408.0	474.0	328.5	366.0
5	524.0	366.5	549.0	283.5	404.5	287.5	519.5	514.5	537.5
6	534.5	441.5	566.0	391.0	399.0	534.0	572.5	559.5	532.0
7	231.0	238.5	342.0	411.0	133.5	126.0	132.0	124.5	100.5
8	404.0	493.0	633.5	480.5	113.5	198.0	378.5	400.0	474.5
9	528.0	447.5	496.0	614.0	322.5	331.0	464.5	468.5	466.0
10	360.0	483.0	458.5	478.5	338.5	313.0	304.0	370.0	325.5
11	413.5	525.5	376.0	602.0	393.5	238.0	370.5	385.0	389.5
12	554.5	468.5	549.0	318.0	502.0	329.5	354.5	505.0	460.0
13	482.5	495.0	508.5	560.0	366.5	306.5	440.5	307.5	307.0
14	398.5	363.0	315.0	406.0	163.5	274.0	378.5	297.0	199.5
15	549.5	520.5	370.5	566.0	185.5	263.5	305.5	398.5	469.0
16	492.0	343.0	402.5	481.5	258.0	299.5	299.0	227.0	313.5
17	569.0	428.5	642.5	567.0	398.5	482.0	422.5	418.0	395.0

Time spent ruminating predicted by rumination model in minutes per hour, as detected by sensor.

ID	-24	-23	-22	-21	-20	-19	-18	-17	-16	-15	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
1	0.0	3.0	34.5	0.0	0.0	29.6	24.0	15.5	7.5	0.0	9.0	2.0	29.1	19.0	17.0	0.0	1.5	12.5	10.5	3.0	6.0	12.5	4.5	0.0
2	22.5	15.4	20.6	34.5	43.9	35.6	12.4	0.0	8.6	22.9	18.0	0.0	30.0	1.5	12.0	0.0	28.5	19.5	33.0	28.5	21.0	17.6	4.9	0.0
3	0.0	0.0	36.7	1.3	3.0	23.5	12.2	33.0	26.0	45.5	16.5	52.5	1.3	32.5	27.2	11.8	20.7	52.0	22.0	26.3	35.5	0.0	10.0	5.0
4	0.0	38.2	6.8	28.5	29.2	17.3	18.0	0.0	39.7	7.5	24.0	50.3	35.2	33.0	2.3	13.5	32.2	54.0	37.5	32.3	12.7	11.3	10.5	3.0
5	1.5	33.0	3.0	0.5	19.5	12.0	0.0	0.0	8.5	0.0	0.0	1.5	1.5	35.0	17.5	5.0	29.5	32.0	29.5	1.5	43.5	29.5	23.0	0.5
6	16.5	3.5	0.5	12.0	26.5	9.0	12.0	15.0	11.0	0.0	1.5	1.5	4.0	9.0	24.0	33.0	35.5	26.0	20.6	18.0	7.5	17.6	5.0	14.0
7	15.6	13.0	0.0	1.5	4.5	0.0	28.5	15.6	16.0	0.0	24.0	21.6	22.0	16.5	13.5	0.0	6.0	10.5	1.5	3.0	6.0	0.6	1.0	0.0
8	0.8	38.7	8.5	36.3	30.5	10.5	2.0	32.5	0.5	23.0	23.0	29.7	20.8	14.0	14.5	32.7	10.5	14.8	25.0	7.0	19.5	20.2	1.3	4.0
9	0.0	29.0	8.6	5.5	6.5	24.5	48.5	35.1	41.0	14.6	16.0	28.5	27.0	47.6	5.0	43.1	26.5	16.1	30.5	1.5	0.5	0.0	0.5	1.5
10	0.0	0.0	0.0	55.2	43.5	38.4	4.0	33.2	22.5	41.0	47.4	1.5	14.5	17.0	10.7	9.5	14.4	35.7	4.4	9.5	6.0	5.5	0.0	0.0
11	16.5	16.2	13.5	20.0	0.0	3.5	0.0	56.3	27.0	31.7	43.5	25.3	59.0	16.7	1.5	52.8	18.7	29.5	22.0	1.0	9.0	0.5	2.0	0.0
12	42.7	36.8	12.7	49.3	35.2	26.5	27.0	27.8	10.5	4.2	3.8	19.5	24.2	51.3	11.0	18.0	21.5	17.7	10.3	28.2	6.0	20.0	13.8	1.5
13	52.5	19.5	40.0	28.0	20.0	26.0	28.0	32.5	21.5	56.5	41.0	12.0	42.5	22.0	58.0	34.0	36.0	12.5	17.5	6.5	7.0	0.0	3.0	1.0
14	12.1	25.4	0.0	6.0	6.0	6.1	15.0	22.4	28.5	27.0	29.5	2.6	26.4	0.0	23.5	32.5	9.1	11.5	19.4	0.0	7.5	0.0	0.0	4.5
15	27.9	13.5	34.5	22.6	33.9	3.1	28.5	0.5	28.0	15.9	22.5	20.6	2.0	0.4	20.6	2.0	3.5	0.5	4.5	0.0	0.5	1.0	0.0	7.5
16	30.0	19.5	6.0	0.0	0.0	27.0	0.5	1.0	35.0	25.0	51.0	4.0	43.0	19.0	42.0	9.0	26.5	40.0	16.5	37.5	9.5	7.0	0.0	1.5
17	0.0	0.0	16.0	0.0	1.0	3.5	16.8	56.2	13.5	26.0	23.5	23.3	45.2	30.3	54.5	29.0	42.7	37.8	31.2	30.3	38.0	55.5	23.2	6.5

Time spent ruminating predicted by rumination model in minutes per hour, as detected by sensor (continued).

ID	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	0.0	0.0	3.0	9.0	0.0	0.0	0.0	2.6	14.5	0.0	0.0	0.0	16.0	0.5	1.0	19.5	5.5	16.0	9.0	21.0	0.0	18.0	5.6	11.0	0.0
2	0.0	1.5	15.0	3.0	0.0	6.0	12.0	19.5	16.5	10.5	0.0	16.5	19.1	15.0	15.4	28.5	1.5	0.0	37.1	19.5	25.9	46.5	1.5	10.1	30.4
3	0.0	0.0	1.5	3.0	1.0	10.5	8.0	0.0	23.0	10.0	14.7	31.0	23.8	1.5	14.7	26.3	9.7	32.3	42.2	8.8	0.0	0.0	14.5	9.2	1.3
4	1.5	0.0	6.0	3.0	15.7	21.8	30.7	17.3	0.0	14.2	28.5	32.3	33.0	6.0	21.0	27.0	24.0	27.0	0.0	24.7	8.3	0.0	0.0	18.7	23.3
5	1.5	4.5	9.0	9.5	6.5	14.5	21.5	34.5	32.5	24.5	14.5	9.5	24.5	28.5	31.5	3.5	16.5	6.5	7.0	18.5	20.5	16.0	9.0	10.5	10.5
6	5.6	7.5	9.5	5.0	2.0	14.1	10.5	47.0	23.0	52.1	4.5	4.6	4.5	36.0	9.5	39.1	10.5	8.0	55.0	34.5	34.0	19.1	20.0	28.1	23.0
7	0.0	0.0	0.0	0.0	1.5	0.0	1.5	0.0	4.5	1.5	0.0	15.0	12.0	0.0	18.0	3.0	1.5	1.5	1.5	1.5	0.0	0.0	3.0	0.0	4.5
8	1.0	8.2	4.3	0.0	11.0	11.0	3.0	0.5	4.5	0.5	5.0	0.0	1.0	14.5	1.0	2.5	15.5	1.0	11.0	0.5	12.5	0.0	2.0	5.7	6.3
9	0.0	0.0	4.0	5.0	22.5	7.5	9.5	8.6	20.5	11.0	9.0	1.5	17.0	15.0	22.5	1.0	6.5	4.0	1.5	0.0	0.0	27.0	3.0	0.0	19.5
10	1.5	5.2	18.0	6.9	21.0	2.7	39.4	9.0	31.0	0.0	40.7	4.9	17.0	17.0	0.0	2.5	0.0	1.0	1.5	5.2	5.9	22.7	13.4	19.5	0.0
11	0.5	1.0	4.0	0.0	18.8	4.7	28.5	39.3	27.0	23.7	0.0	17.5	27.8	8.7	19.3	14.2	3.0	12.8	2.2	0.5	10.0	0.0	0.5	1.0	1.5
12	0.0	0.7	39.5	18.8	12.0	18.0	4.0	6.5	16.5	1.0	14.0	24.2	13.8	26.7	17.3	12.0	2.0	19.7	1.8	9.5	17.5	25.0	8.5	0.0	25.5
13	0.0	0.0	6.0	3.5	2.0	21.0	7.5	7.0	24.0	0.5	32.0	14.0	18.0	10.5	1.0	21.5	11.5	17.5	3.0	15.0	1.5	1.5	10.0	2.5	1.0
14	0.0	0.5	1.0	0.0	4.5	0.0	4.0	4.0	5.5	6.1	1.4	0.0	21.1	10.4	7.5	15.6	8.9	4.5	1.5	0.0	0.0	4.5	17.5	12.0	13.5
15	0.0	2.5	12.0	0.0	12.0	11.5	11.5	15.5	26.9	23.1	0.5	10.4	14.6	8.9	9.1	0.0	19.9	18.0	9.6	7.5	14.5	0.0	6.4	24.5	4.1
16	0.0	0.0	1.0	0.0	0.0	0.0	0.5	10.0	0.0	3.0	0.5	16.5	24.0	44.0	8.5	22.0	25.5	0.0	14.5	22.5	15.0	0.0	0.0	0.0	14.0
17	0.0	5.5	0.0	1.0	7.0	0.0	22.0	3.0	5.0	3.0	17.3	9.7	5.8	15.7	5.5	33.8	45.0	6.7	24.3	30.7	3.5	30.3	24.5	39.0	43.5

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